

# FINAL PRESENTATION DSC

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# complete sheet

complete ecommerce spreadsheet

| Obs | Date       | Product_Category | Price  | Discount | Customer_Segment | Marketing_Spend | Units_Sold |
|-----|------------|------------------|--------|----------|------------------|-----------------|------------|
| 1   | 01/01/2023 | Sports           | 932.8  | 35.82    | Occasional       | 6780.38         | 32         |
| 2   | 02/01/2023 | Toys             | 569.48 | 3.6      | Premium          | 6807.56         | 16         |
| 3   | 03/01/2023 | Home Decor       | 699.68 | 3.56     | Premium          | 3793.91         | 27         |
| 4   | 04/01/2023 | Toys             | 923.27 | 0.61     | Premium          | 9422.75         | 29         |
| 5   | 05/01/2023 | Toys             | 710.17 | 47.83    | Premium          | 1756.83         | 17         |
| 6   | 06/01/2023 | Fashion          | 161.01 | 36.88    | Premium          | 5053.56         | 27         |
| 7   | 07/01/2023 | Home Decor       | 580.53 | 17.66    | Occasional       | 6939.75         | 30         |
| 8   | 08/01/2023 | Home Decor       | 610.65 | 14.83    | Regular          | 7001.64         | 27         |
| 9   | 09/01/2023 | Home Decor       | 429.89 | 17.49    | Occasional       | 6521.53         | 32         |
| 10  | 10/01/2023 | Toys             | 739.08 | 38.73    | Regular          | 2825.35         | 28         |
| 11  | 11/01/2023 | Sports           | 935.02 | 33.07    | Premium          | 1646.45         | 26         |
| 12  | 12/01/2023 | Home Decor       | 926.31 | 9.26     | Regular          | 6395.81         | 26         |
| 13  | 13/01/2023 | Toys             | 456.33 | 8.71     | Occasional       | 6033.09         | 37         |
| 14  | 14/01/2023 | Fashion          | 122.11 | 4.92     | Occasional       | 1875.62         | 28         |

|      |            |             |        |       |            |         |    |
|------|------------|-------------|--------|-------|------------|---------|----|
| 984  | 10/09/2025 | Fashion     | 842.46 | 10.46 | Occasional | 820.08  | 23 |
| 985  | 11/09/2025 | Home Decor  | 48.04  | 2.5   | Regular    | 2877.95 | 21 |
| 986  | 12/09/2025 | Home Decor  | 902.74 | 42.18 | Occasional | 3564.38 | 36 |
| 987  | 13/09/2025 | Toys        | 466.86 | 49.06 | Occasional | 7846.02 | 23 |
| 988  | 14/09/2025 | Sports      | 640.83 | 39.66 | Regular    | 9925.89 | 34 |
| 989  | 15/09/2025 | Home Decor  | 662.76 | 42.69 | Occasional | 2482.64 | 21 |
| 990  | 16/09/2025 | Fashion     | 896.17 | 12.1  | Regular    | 8755.64 | 23 |
| 991  | 17/09/2025 | Fashion     | 640.3  | 48.03 | Premium    | 8323.17 | 33 |
| 992  | 18/09/2025 | Home Decor  | 617.79 | 9.85  | Premium    | 2324.11 | 26 |
| 993  | 19/09/2025 | Sports      | 75.99  | 47.57 | Regular    | 4053.39 | 42 |
| 994  | 20/09/2025 | Fashion     | 523.22 | 49.74 | Premium    | 4155.76 | 35 |
| 995  | 21/09/2025 | Toys        | 158.67 | 35.59 | Premium    | 9784.38 | 26 |
| 996  | 22/09/2025 | Fashion     | 740.06 | 49.06 | Regular    | 1890.83 | 34 |
| 997  | 23/09/2025 | Electronics | 517.1  | 28.48 | Premium    | 8011.66 | 21 |
| 998  | 24/09/2025 | Electronics | 683.43 | 12.98 | Occasional | 3408.51 | 19 |
| 999  | 25/09/2025 | Sports      | 51.26  | 21.85 | Occasional | 7335.62 | 26 |
| 1000 | 26/09/2025 | Home Decor  | 93.94  | 29.68 | Regular    | 4258.11 | 39 |

It contains 1000 obs

With 7 variables and unit sold is the dependant variable (Y) where as other variable are independent variable (X)

# DUMMY VARIABLE

Spreadsheet with dummy variable

| Obs | Date       | Product_Category | Price  | Discount | Customer_Segment | Marketing_Spend | Units_Sold | D_Premium | D_Regular | D_Sports | D_Toys | D_Fashion | D_HomeDecor |
|-----|------------|------------------|--------|----------|------------------|-----------------|------------|-----------|-----------|----------|--------|-----------|-------------|
| 1   | 01/01/2023 | Sports           | 932.8  | 35.82    | Occasional       | 6780.38         | 32         | 0         | 0         | 1        | 0      | 0         | 0           |
| 2   | 02/01/2023 | Toys             | 569.48 | 3.6      | Premium          | 6807.56         | 16         | 1         | 0         | 0        | 1      | 0         | 0           |
| 3   | 03/01/2023 | Home Decor       | 699.68 | 3.56     | Premium          | 3793.91         | 27         | 1         | 0         | 0        | 0      | 0         | 1           |
| 4   | 04/01/2023 | Toys             | 923.27 | 0.61     | Premium          | 9422.75         | 29         | 1         | 0         | 0        | 1      | 0         | 0           |
| 5   | 05/01/2023 | Toys             | 710.17 | 47.83    | Premium          | 1756.83         | 17         | 1         | 0         | 0        | 1      | 0         | 0           |
| 6   | 06/01/2023 | Fashion          | 161.01 | 36.88    | Premium          | 5053.56         | 27         | 1         | 0         | 0        | 0      | 1         | 0           |
| 7   | 07/01/2023 | Home Decor       | 580.53 | 17.66    | Occasional       | 6939.75         | 30         | 0         | 0         | 0        | 0      | 0         | 1           |
| 8   | 08/01/2023 | Home Decor       | 610.65 | 14.83    | Regular          | 7001.64         | 27         | 0         | 1         | 0        | 0      | 0         | 1           |
| 9   | 09/01/2023 | Home Decor       | 429.89 | 17.49    | Occasional       | 6521.53         | 32         | 0         | 0         | 0        | 0      | 0         | 1           |
| 10  | 10/01/2023 | Toys             | 739.08 | 38.73    | Regular          | 2825.35         | 28         | 0         | 1         | 0        | 1      | 0         | 0           |
| 11  | 11/01/2023 | Sports           | 935.02 | 33.07    | Premium          | 1646.45         | 26         | 1         | 0         | 1        | 0      | 0         | 0           |
| 12  | 12/01/2023 | Home Decor       | 926.31 | 9.26     | Regular          | 6395.81         | 26         | 0         | 1         | 0        | 0      | 0         | 1           |
| 13  | 13/01/2023 | Toys             | 456.33 | 8.71     | Occasional       | 6033.09         | 37         | 0         | 0         | 0        | 1      | 0         | 0           |
| 14  | 14/01/2023 | Fashion          | 122.11 | 4.92     | Occasional       | 1875.62         | 28         | 0         | 0         | 0        | 0      | 1         | 0           |
| 15  | 15/01/2023 | Sports           | 984.99 | 33.02    | Regular          | 7080.88         | 27         | 0         | 1         | 1        | 0      | 0         | 0           |
| 16  | 16/01/2023 | Fashion          | 840.51 | 38.22    | Premium          | 4606.2          | 32         | 1         | 0         | 0        | 0      | 1         | 0           |
| 17  | 17/01/2023 | Sports           | 133.42 | 13.25    | Premium          | 6710.83         | 20         | 1         | 0         | 1        | 0      | 0         | 0           |
| 18  | 18/01/2023 | Toys             | 921.63 | 1.05     | Occasional       | 8389.93         | 35         | 0         | 0         | 0        | 1      | 0         | 0           |
| 19  | 19/01/2023 | Electronics      | 871.2  | 4.11     | Regular          | 1780.31         | 23         | 0         | 1         | 0        | 0      | 0         | 0           |
| 20  | 20/01/2023 | Sports           | 523.65 | 48.39    | Premium          | 289.53          | 17         | 1         | 0         | 1        | 0      | 0         | 0           |
| 21  | 21/01/2023 | Fashion          | 595.36 | 14.77    | Premium          | 7813.12         | 23         | 1         | 0         | 0        | 0      | 1         | 0           |
| 22  | 22/01/2023 | Toys             | 405.01 | 38.46    | Occasional       | 6136            | 24         | 0         | 0         | 0        | 1      | 0         | 0           |
| 23  | 23/01/2023 | Sports           | 64.21  | 31.23    | Premium          | 7026.43         | 35         | 1         | 0         | 1        | 0      | 0         | 0           |
| 24  | 24/01/2023 | Electronics      | 341.85 | 19.1     | Regular          | 8397.73         | 32         | 0         | 1         | 0        | 0      | 0         | 0           |
| 25  | 25/01/2023 | Electronics      | 804.82 | 10.28    | Occasional       | 8047.83         | 34         | 0         | 0         | 0        | 0      | 0         | 0           |
| 26  | 26/01/2023 | Home Decor       | 14.59  | 6.07     | Occasional       | 9613.11         | 29         | 0         | 0         | 0        | 0      | 0         | 1           |
| 27  | 27/01/2023 | Home Decor       | 340.16 | 30.75    | Regular          | 5405.76         | 25         | 0         | 1         | 0        | 0      | 0         | 1           |
| 28  | 28/01/2023 | Fashion          | 404.19 | 38.73    | Occasional       | 4936.11         | 29         | 0         | 0         | 0        | 0      | 1         | 0           |
| 29  | 29/01/2023 | Sports           | 542.02 | 32.2     | Premium          | 4078.68         | 29         | 1         | 0         | 1        | 0      | 0         | 0           |

SAS can't use text data in regression, so I changed the category names into numbers using dummy variables. Each dummy variable shows 1 if that category is present and 0 if not.

## Summary Statistics of Numeric Variables

### The MEANS Procedure

| Variable        | N    | Mean        | Std Dev     | Minimum     | Maximum     |
|-----------------|------|-------------|-------------|-------------|-------------|
| Price           | 1000 | 505.2419200 | 289.7912481 | 14.5900000  | 999.4200000 |
| Discount        | 1000 | 24.9197500  | 14.4399513  | 0.0100000   | 49.9200000  |
| Marketing_Spend | 1000 | 4912.83     | 2840.60     | 100.3000000 | 9995.62     |
| Units_Sold      | 1000 | 29.6310000  | 7.2679889   | 5.0000000   | 57.0000000  |
| D_Premium       | 1000 | 0.3160000   | 0.4651455   | 0           | 1.0000000   |
| D_Regular       | 1000 | 0.3450000   | 0.4756061   | 0           | 1.0000000   |
| D_Sports        | 1000 | 0.2060000   | 0.4046328   | 0           | 1.0000000   |
| D_Toys          | 1000 | 0.2040000   | 0.4031706   | 0           | 1.0000000   |
| D_Fashion       | 1000 | 0.1900000   | 0.3924972   | 0           | 1.0000000   |
| D_HomeDecor     | 1000 | 0.1900000   | 0.3924972   | 0           | 1.0000000   |

Standard deviation shows how much the numbers go up and down from the average. The average price is around \$505, but the standard deviation is very high (about \$290), which means there's a big difference between cheaper and expensive products.

The average discount is about 25%, and it also varies a lot.

This shows that some products are heavily discounted, while others are sold at full price.

## Distribution of Units Sold

### The UNIVARIATE Procedure

| Moments         |            |                  |            |
|-----------------|------------|------------------|------------|
| N               | 1000       | Sum Weights      | 1000       |
| Mean            | 29.631     | Sum Observations | 29631      |
| Std Deviation   | 7.2679889  | Variance         | 52.8236627 |
| Skewness        | 0.11861553 | Kurtosis         | -0.0579639 |
| Uncorrected SS  | 930767     | Corrected SS     | 52770.839  |
| Coeff Variation | 24.5283281 | Std Error Mean   | 0.22983399 |

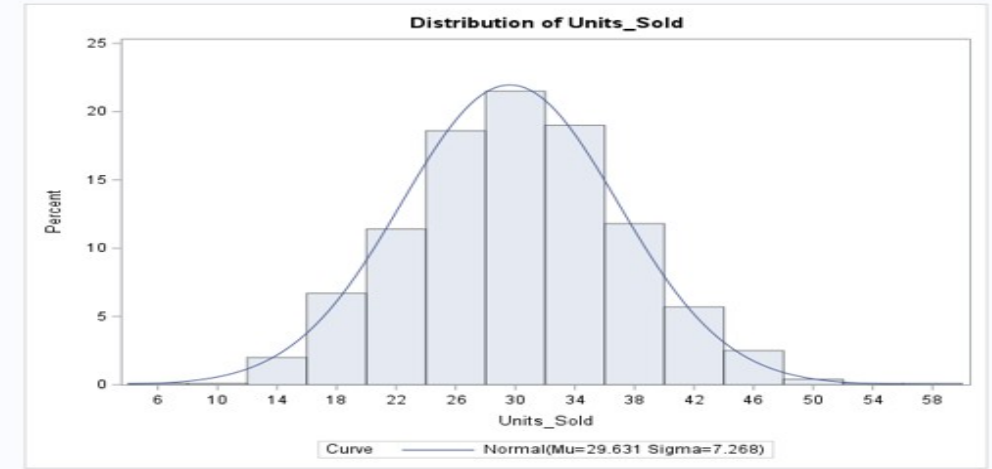
| Basic Statistical Measures |          |                     |          |
|----------------------------|----------|---------------------|----------|
| Location                   |          | Variability         |          |
| Mean                       | 29.63100 | Std Deviation       | 7.26799  |
| Median                     | 30.00000 | Variance            | 52.82366 |
| Mode                       | 31.00000 | Range               | 52.00000 |
|                            |          | Interquartile Range | 10.00000 |

| Tests for Location: Mu0=0 |            |          |        |
|---------------------------|------------|----------|--------|
| Test                      | Statistic  | p Value  |        |
| Student's t               | t 128.9235 | Pr >  t  | <.0001 |
| Sign                      | M 500      | Pr >=  M | <.0001 |
| Signed Rank               | S 250250   | Pr >=  S | <.0001 |

| Tests for Normality |           |          |           |         |
|---------------------|-----------|----------|-----------|---------|
| Test                | Statistic | p Value  |           |         |
| Shapiro-Wilk        | W         | 0.996503 | Pr < W    | 0.0251  |
| Kolmogorov-Smirnov  | D         | 0.03478  | Pr > D    | <0.0100 |
| Cramer-von Mises    | W-Sq      | 0.16955  | Pr > W-Sq | 0.0142  |
| Anderson-Darling    | A-Sq      | 1.008981 | Pr > A-Sq | 0.0123  |

## Distribution of Units Sold

### The UNIVARIATE Procedure



This histogram clearly shows a normal, bell-shaped curve centered around 30. Most products sell between 20 and 40 units.

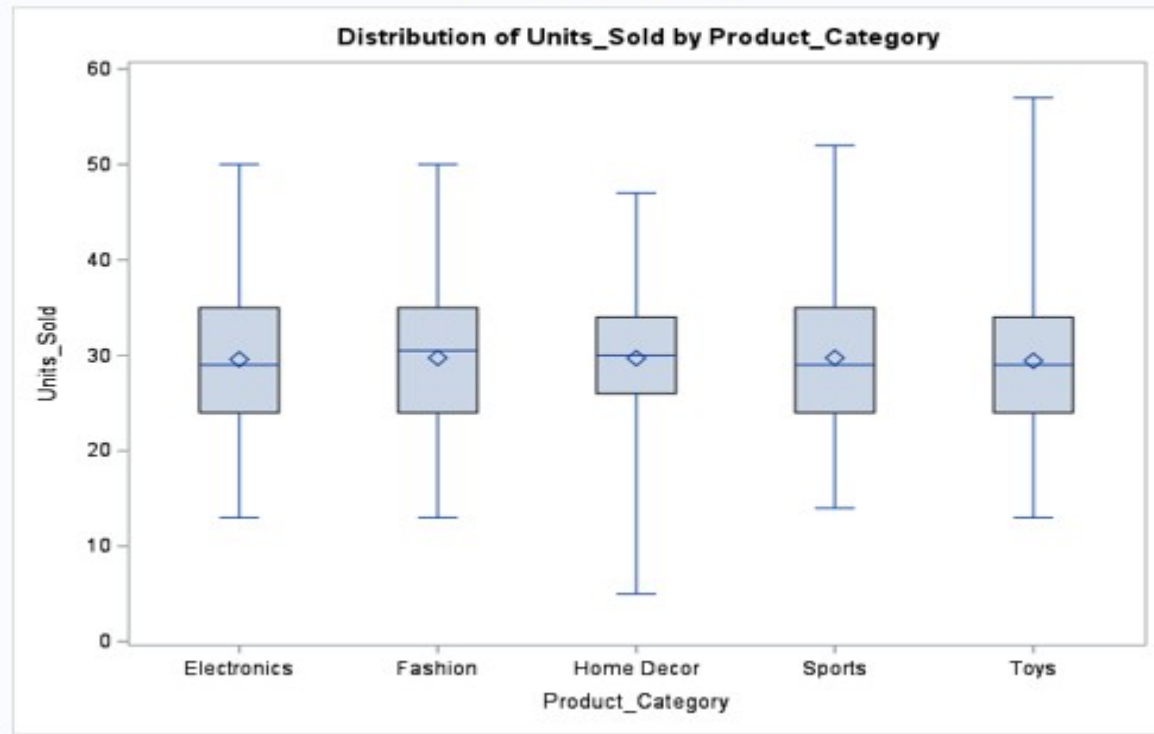
Before building my regression model, I first checked how my main variable (*Units Sold*) is distributed. This helps me see whether sales are spread normally or have extreme high or extreme lows.

The values of mean, median, and mode are very close (29.6, 30, 31), which means the data is well-balanced.

The standard deviation (7.26) shows sales don't change a lot between products — most stay near the average.

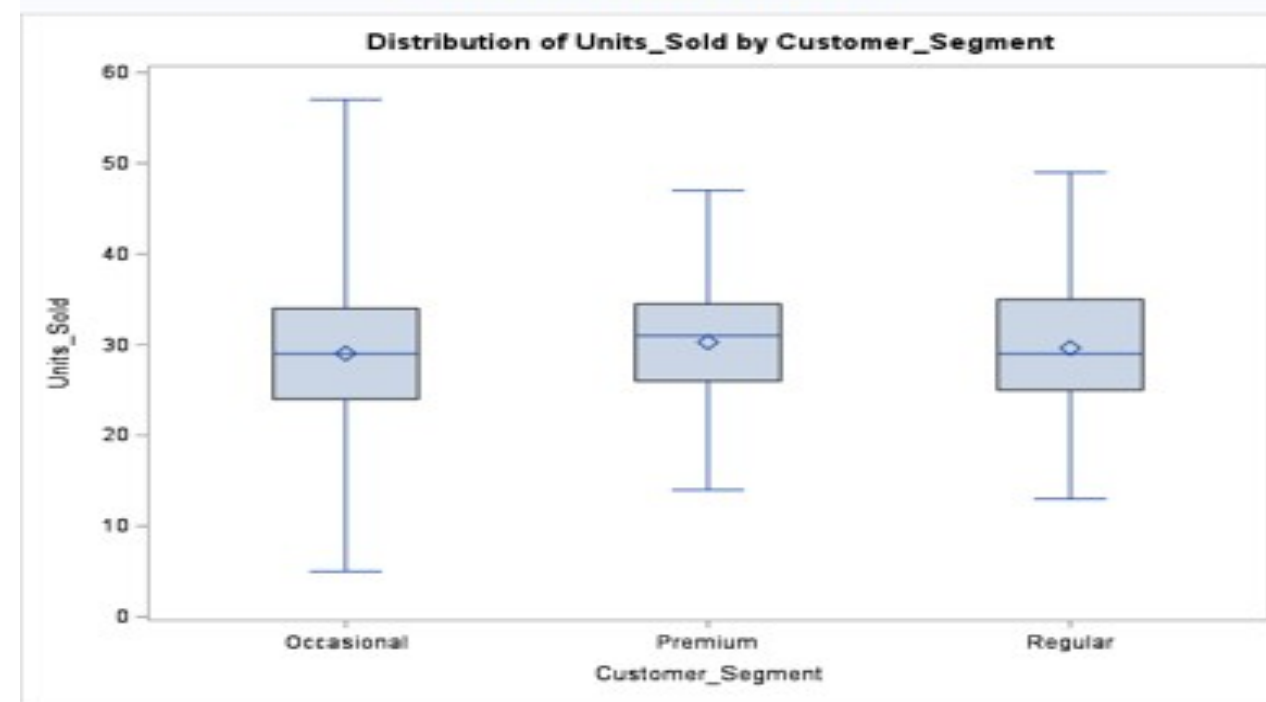
The skewness and kurtosis are both near zero, which means the data follows a normal bell shape."

Boxplot(by Product Category)



The sales for all product categories look almost the same, which means they are quite consistent. Fashion products have a little higher average and middle value, so they sell a bit more than the others. Home Decor and Toys have slightly lower sales, but not by much. There are no extreme values or unusual points, so the data looks clean and balanced.

Boxplot(by customer segment)



Among the customer segments, Premium customers buy the most on average, shown by their higher middle value. Regular customers are in between, while Occasional customers have more ups and downs, sometimes they buy a lot, and sometimes very little.



### Correlation of Units Sold with All Predictor Variables

#### The CORR Procedure

10 Variables: Units\_Sold Price Discount Marketing\_Spend D\_Premium D\_Regular D\_Sports D\_Toys D\_Fashion D\_HomeDecor

| Simple Statistics |      |           |           |           |           |           |
|-------------------|------|-----------|-----------|-----------|-----------|-----------|
| Variable          | N    | Mean      | Std Dev   | Sum       | Minimum   | Maximum   |
| Units_Sold        | 1000 | 29.63100  | 7.26799   | 29631     | 5.00000   | 57.00000  |
| Price             | 1000 | 505.24192 | 289.79125 | 505242    | 14.59000  | 999.42000 |
| Discount          | 1000 | 24.91975  | 14.43995  | 24920     | 0.01000   | 49.92000  |
| Marketing_Spend   | 1000 | 4913      | 2841      | 4912830   | 100.30000 | 9996      |
| D_Premium         | 1000 | 0.31600   | 0.46515   | 316.00000 | 0         | 1.00000   |
| D_Regular         | 1000 | 0.34500   | 0.47561   | 345.00000 | 0         | 1.00000   |
| D_Sports          | 1000 | 0.20600   | 0.40463   | 206.00000 | 0         | 1.00000   |
| D_Toys            | 1000 | 0.20400   | 0.40317   | 204.00000 | 0         | 1.00000   |
| D_Fashion         | 1000 | 0.19000   | 0.39250   | 190.00000 | 0         | 1.00000   |
| D_HomeDecor       | 1000 | 0.19000   | 0.39250   | 190.00000 | 0         | 1.00000   |

#### Pearson Correlation Coefficients, N = 1000 Prob > |r| under H0: Rho=0

|                 | Units_Sold         | Price              | Discount           | Marketing_Spend    | D_Premium          | D_Regular          | D_Sports           | D_Toys             | D_Fashion          | D_HomeDecor        |
|-----------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| Units_Sold      | 1.00000            | 0.01505<br>0.6346  | -0.03249<br>0.3046 | 0.00947<br>0.7648  | 0.06029<br>0.0567  | -0.00020<br>0.9949 | 0.00715<br>0.8213  | -0.01425<br>0.6526 | 0.00706<br>0.8236  | 0.00460<br>0.8845  |
| Price           | 0.01505<br>0.6346  | 1.00000            | 0.03077<br>0.3311  | -0.00290<br>0.9269 | -0.03333<br>0.2923 | -0.01693<br>0.5929 | 0.03679<br>0.2450  | -0.01357<br>0.6683 | 0.02879<br>0.3631  | -0.02181<br>0.4908 |
| Discount        | -0.03249<br>0.3046 | 0.03077<br>0.3311  | 1.00000            | 0.01081<br>0.7327  | -0.02007<br>0.5261 | 0.03166<br>0.3173  | -0.03110<br>0.3259 | 0.05856<br>0.0642  | 0.05730<br>0.0701  | -0.09337<br>0.0031 |
| Marketing_Spend | 0.00947<br>0.7648  | -0.00290<br>0.9269 | 0.01081<br>0.7327  | 1.00000            | -0.01171<br>0.7115 | 0.01912<br>0.5460  | -0.02943<br>0.3525 | 0.00728<br>0.8181  | -0.01121<br>0.7234 | 0.00475<br>0.8808  |
| D_Premium       | 0.06029<br>0.0567  | -0.03333<br>0.2923 | -0.02007<br>0.5261 | -0.01171<br>0.7115 | 1.00000            | -0.49329<br><.0001 | 0.01544<br>0.6257  | 0.01354<br>0.6690  | -0.00022<br>0.9945 | -0.00570<br>0.8571 |
| D_Regular       | -0.00020<br>0.9949 | -0.01693<br>0.5929 | 0.03166<br>0.3173  | 0.01912<br>0.5460  | -0.49329<br><.0001 | 1.00000            | 0.02564<br>0.4179  | -0.04375<br>0.1669 | -0.01367<br>0.6658 | -0.01904<br>0.5477 |
| D_Sports        | 0.00715<br>0.8213  | 0.03679<br>0.2450  | -0.03110<br>0.3259 | -0.02943<br>0.3525 | 0.01544<br>0.6257  | 0.02564<br>0.4179  | 1.00000            | -0.25786<br><.0001 | -0.24669<br><.0001 | -0.24669<br><.0001 |
| D_Toys          | -0.01425<br>0.6526 | -0.01357<br>0.6683 | 0.05856<br>0.0642  | 0.00728<br>0.8181  | 0.01354<br>0.6690  | -0.04375<br>0.1669 | -0.25786<br><.0001 | 1.00000            | -0.24518<br><.0001 | -0.24518<br><.0001 |
| D_Fashion       | 0.00706<br>0.8236  | 0.02879<br>0.3631  | 0.05730<br>0.0701  | -0.01121<br>0.7234 | -0.00022<br>0.9945 | -0.01367<br>0.6658 | -0.24669<br><.0001 | -0.24518<br><.0001 | 1.00000            | -0.23457<br><.0001 |
| D_HomeDecor     | 0.00460<br>0.8845  | -0.02181<br>0.4908 | -0.09337<br>0.0031 | 0.00475<br>0.8808  | -0.00570<br>0.8571 | -0.01904<br>0.5477 | -0.24669<br><.0001 | -0.24518<br><.0001 | -0.23457<br><.0001 | 1.00000            |

The correlations with Units Sold are Price (0.015), Discount (-0.032), and Marketing Spend (0.009) — all very close to zero. This means that none of these variables have a strong direct relationship with sales.

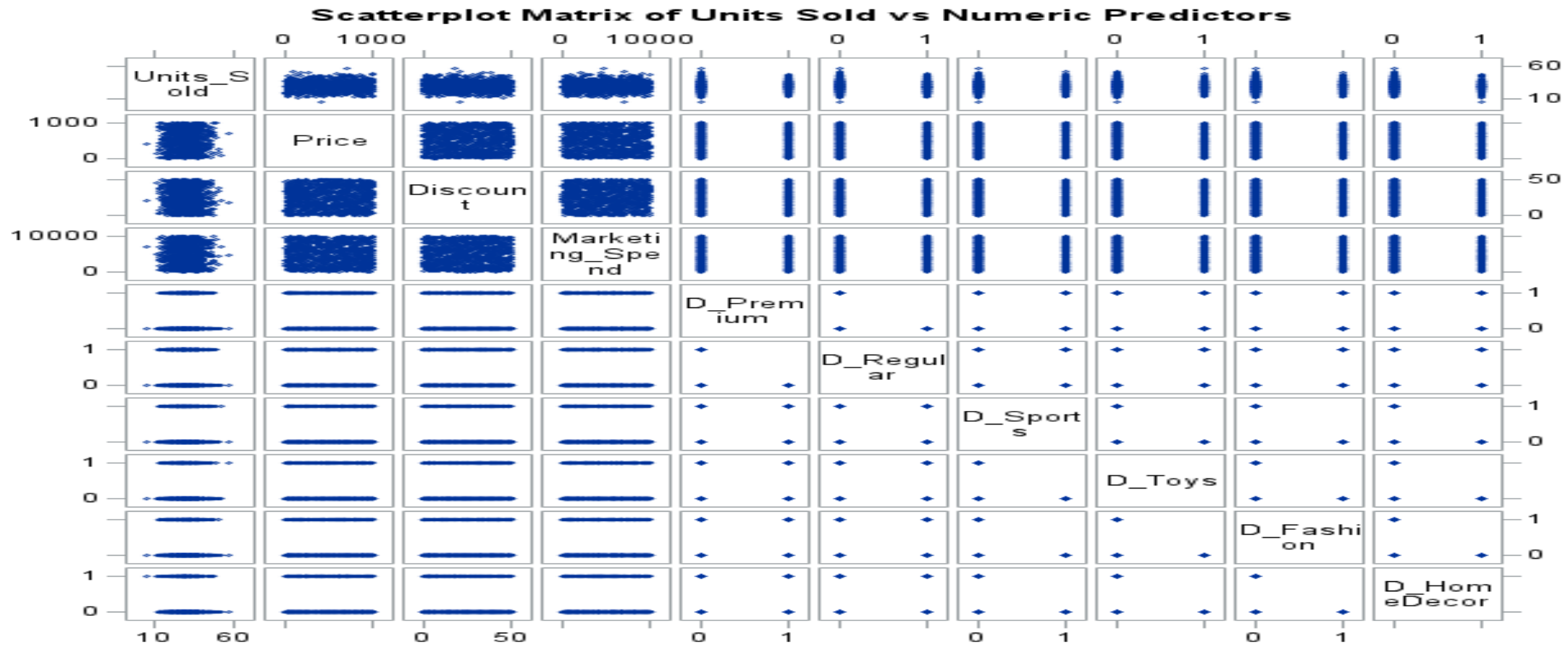
D\_Premium (0.06) shows a small positive link with sales — Premium customers buy slightly more.

D\_Regular (-0.00) has no correlation.

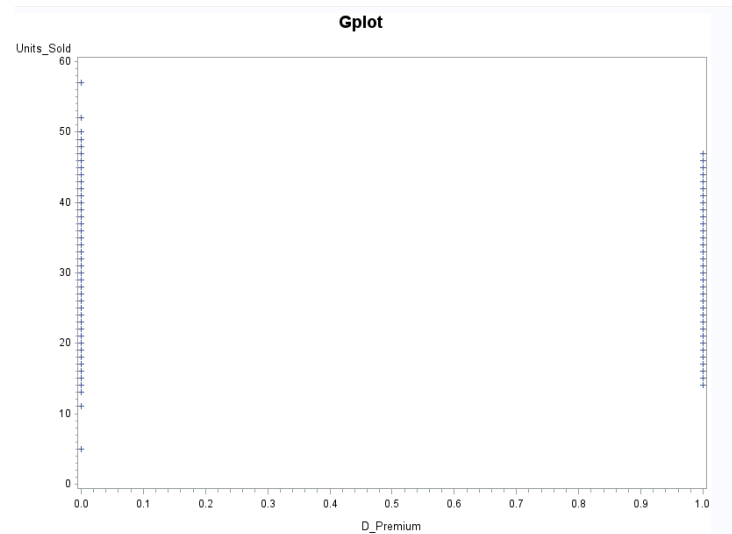
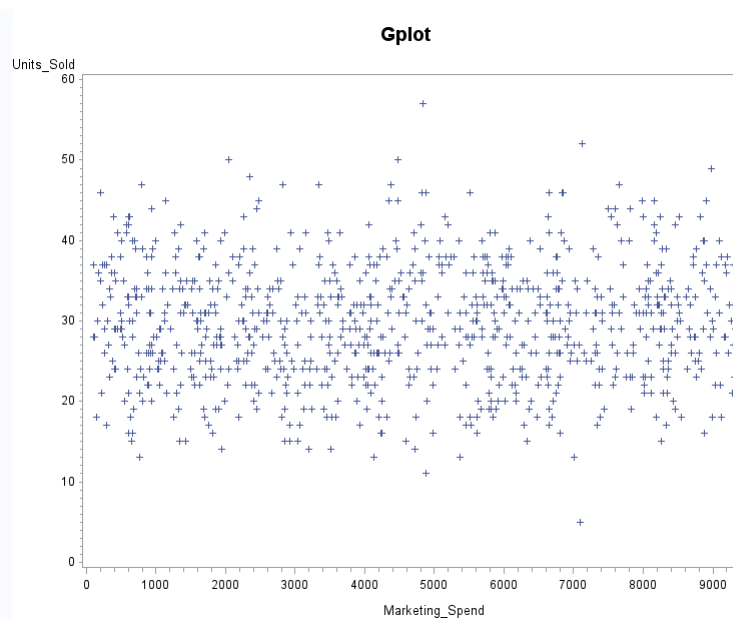
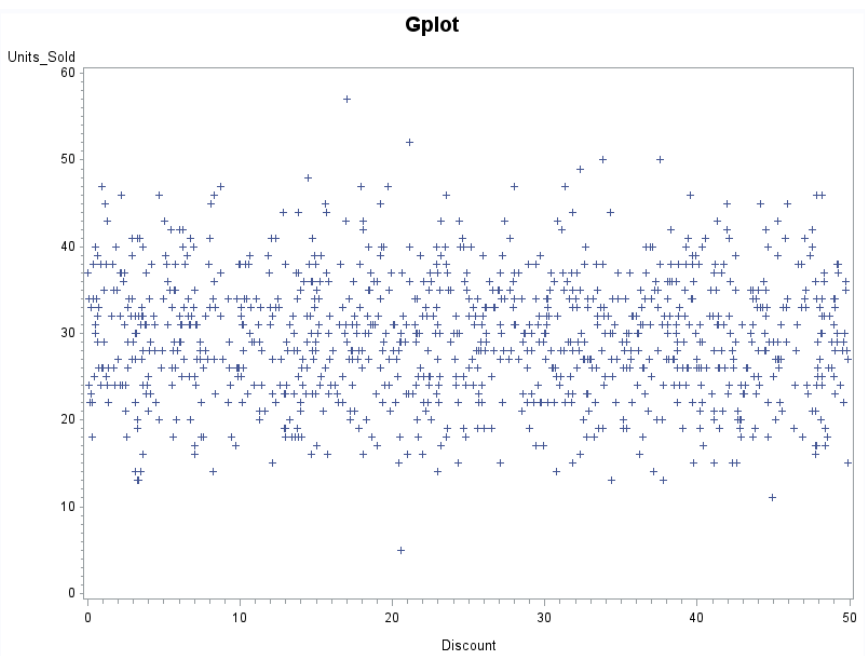
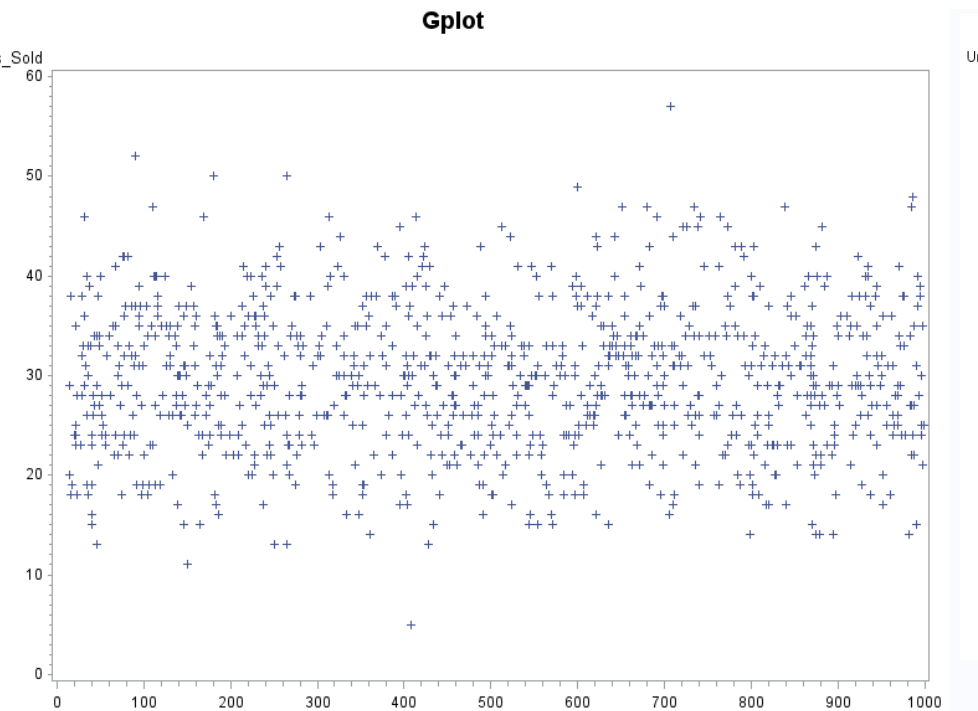
Premium customers contribute a bit more to total sales, while Regular ones show no special trend.

All categories (Sports, Toys, Fashion, Home Decor) have weak or near-zero correlations.

This means product category alone doesn't affect sales much



This scatterplot matrix shows how Units Sold relates to other variables like Price, Discount, and Marketing Spend. Each small box compares two variables to check if there's any clear pattern. The points are spread out randomly, meaning there is no strong linear relationship between sales and these predictors. The dummy variables, such as D\_Premium or D\_Sports, show two straight lines because they only take values 0 or 1, which is normal. Overall, the plots show that the data is spread evenly without strong outliers. This means Units Sold is not strongly affected by any single



(ALL DUMMY VARIABLES  
ARE LIKE THIS)

The scatterplots for Price, Discount, and Marketing Spend don't show any clear pattern going up or down. The points are spread randomly, meaning sales don't strongly rise or fall with these factors. For dummy variables like D\_Premium, D\_Regular, or D\_Sports, there are two straight vertical lines because these variables only have 0 or 1 values — this is normal for categories. The overall spread looks even, and there are no extreme outliers in the data.



# This is my full multiple regression model. It includes all predictors

Full Multiple Regression Model

The REG Procedure  
Model: MODEL1  
Dependent Variable: Units\_Sold

|                             |      |
|-----------------------------|------|
| Number of Observations Read | 1000 |
| Number of Observations Used | 1000 |

Analysis of Variance

| Source          | DF  | Sum of Squares | Mean Square | F Value | Pr > F |
|-----------------|-----|----------------|-------------|---------|--------|
| Model           | 9   | 342.92880      | 38.10320    | 0.72    | 0.6914 |
| Error           | 990 | 52428          | 52.95749    |         |        |
| Corrected Total | 999 | 52771          |             |         |        |

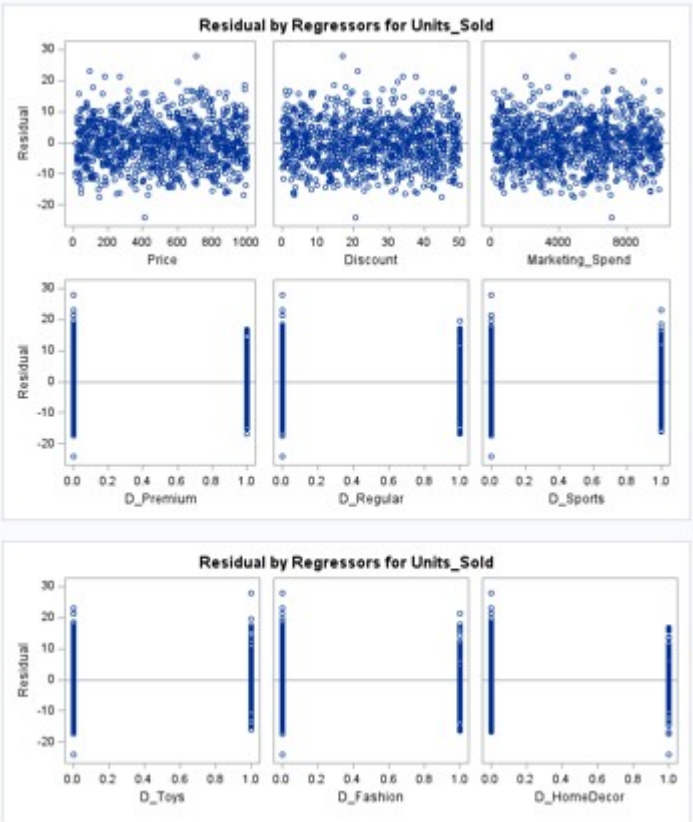
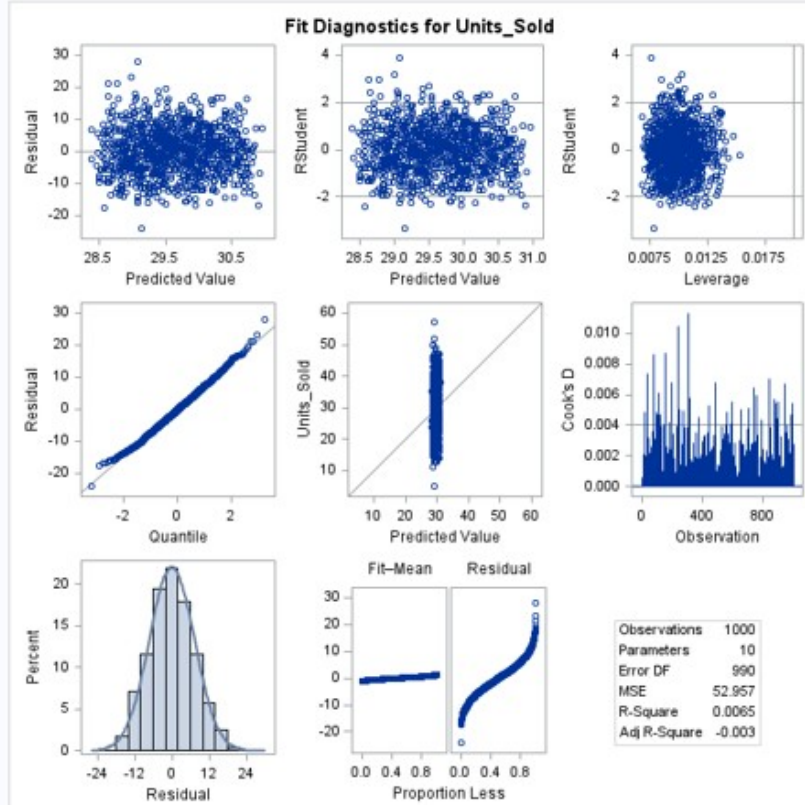
|                |          |          |         |
|----------------|----------|----------|---------|
| Root MSE       | 7.27719  | R-Square | 0.0065  |
| Dependent Mean | 29.63100 | Adj R-Sq | -0.0025 |
| Coeff Var      | 24.55938 |          |         |

Parameter Estimates

| Variable        | DF | Parameter Estimate | Standard Error | t Value | Pr >  t |
|-----------------|----|--------------------|----------------|---------|---------|
| Intercept       | 1  | 29.00284           | 0.91763        | 31.61   | <.0001  |
| Price           | 1  | 0.00047564         | 0.00079713     | 0.60    | 0.5509  |
| Discount        | 1  | -0.01630           | 0.01607        | -1.01   | 0.3105  |
| Marketing_Spend | 1  | 0.00002648         | 0.00008113     | 0.33    | 0.7442  |
| D_Premium       | 1  | 1.25551            | 0.57007        | 2.20    | 0.0279  |
| D_Regular       | 1  | 0.61756            | 0.55858        | 1.11    | 0.2692  |
| D_Sports        | 1  | 0.10379            | 0.71517        | 0.15    | 0.8846  |
| D_Toys          | 1  | -0.11071           | 0.71713        | -0.15   | 0.8773  |
| D_Fashion       | 1  | 0.19001            | 0.73029        | 0.26    | 0.7948  |
| D_HomeDecor     | 1  | 0.10174            | 0.73094        | 0.14    | 0.8893  |

Full Multiple Regression Model

The REG Procedure  
Model: MODEL1  
Dependent Variable: Units\_Sold



From the table, the F-value is 0.72 and the p-value is 0.69, which means the model is not statistically significant. The R-square is 0.0065, so the model explains only about 0.65% of the total variation in Units Sold. That's a very small amount. The Adjusted R-square is negative, which means adding more predictors didn't help — it actually made the model a bit worse. The Root MSE is around 7.3, so the model's average prediction error is about 7 units. Only the Premium customer variable shows a small positive effect on sales.

Now looking at the diagnostic graphs: The residual vs predicted plot shows random scatter — that's good, it means the errors are independent. Most studentized residuals are between -2 and +2, showing no extreme outliers. The Q-Q plot and histogram both look roughly normal, so residuals follow a normal pattern. The leverage and Cook's D plots show no points with strong influence.

## Full Regression Model with Multicollinearity Diagnostics

The REG Procedure  
Model: MODEL1  
Dependent Variable: Units\_Sold

|                             |      |
|-----------------------------|------|
| Number of Observations Read | 1000 |
| Number of Observations Used | 1000 |

| Analysis of Variance |     |                |             |         |        |
|----------------------|-----|----------------|-------------|---------|--------|
| Source               | DF  | Sum of Squares | Mean Square | F Value | Pr > F |
| Model                | 9   | 342.92880      | 38.10320    | 0.72    | 0.6914 |
| Error                | 990 | 52428          | 52.95749    |         |        |
| Corrected Total      | 999 | 52771          |             |         |        |

|                |          |          |         |
|----------------|----------|----------|---------|
| Root MSE       | 7.27719  | R-Square | 0.0065  |
| Dependent Mean | 29.63100 | Adj R-Sq | -0.0025 |
| Coeff Var      | 24.55938 |          |         |

| Parameter Estimates |    |                    |                |         |         |                    |
|---------------------|----|--------------------|----------------|---------|---------|--------------------|
| Variable            | DF | Parameter Estimate | Standard Error | t Value | Pr >  t | Variance Inflation |
| Intercept           | 1  | 29.00284           | 0.91763        | 31.61   | <.0001  | 0                  |
| Price               | 1  | 0.00047564         | 0.00079713     | 0.60    | 0.5509  | 1.00663            |
| Discount            | 1  | -0.01630           | 0.01607        | -1.01   | 0.3105  | 1.01563            |
| Marketing_Spend     | 1  | 0.00002648         | 0.00008113     | 0.33    | 0.7442  | 1.00194            |
| D_Premium           | 1  | 1.25551            | 0.57007        | 2.20    | 0.0279  | 1.32640            |
| D_Regular           | 1  | 0.61756            | 0.55858        | 1.11    | 0.2692  | 1.33139            |
| D_Sports            | 1  | 0.10379            | 0.71517        | 0.15    | 0.8846  | 1.57971            |
| D_Toys              | 1  | -0.11071           | 0.71713        | -0.15   | 0.8773  | 1.57694            |
| D_Fashion           | 1  | 0.19001            | 0.73029        | 0.26    | 0.7948  | 1.54990            |
| D_HomeDecor         | 1  | 0.10174            | 0.73094        | 0.14    | 0.8893  | 1.55265            |

Here we can see the full regression model with multicollinearity check. Multicollinearity means when two or more predictors are very closely related, which makes it hard to tell which one really affects the result. In my model, all the VIF values are between 1 and 1.8, which are very low. This means there is no multicollinearity — my variables are independent and not affecting each other.

Regression Model Diagnostics – Outliers & Influential Points

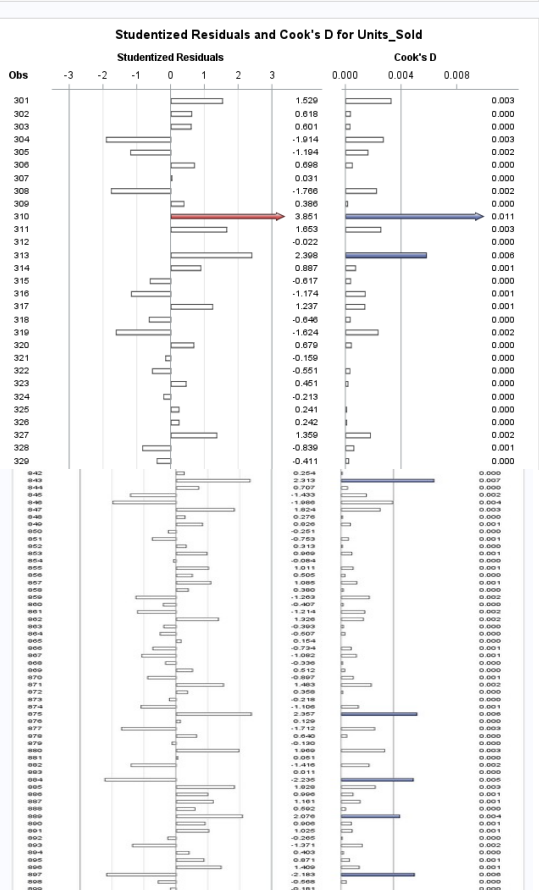
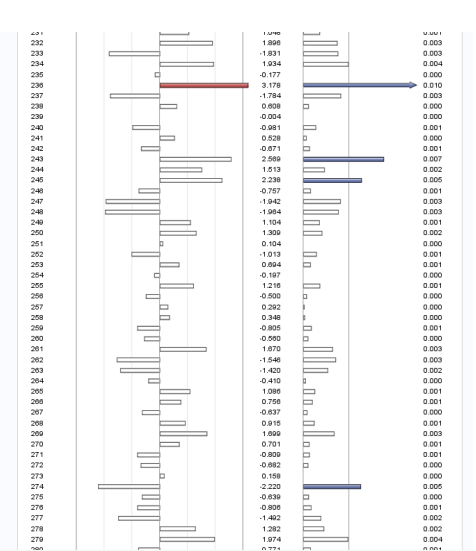
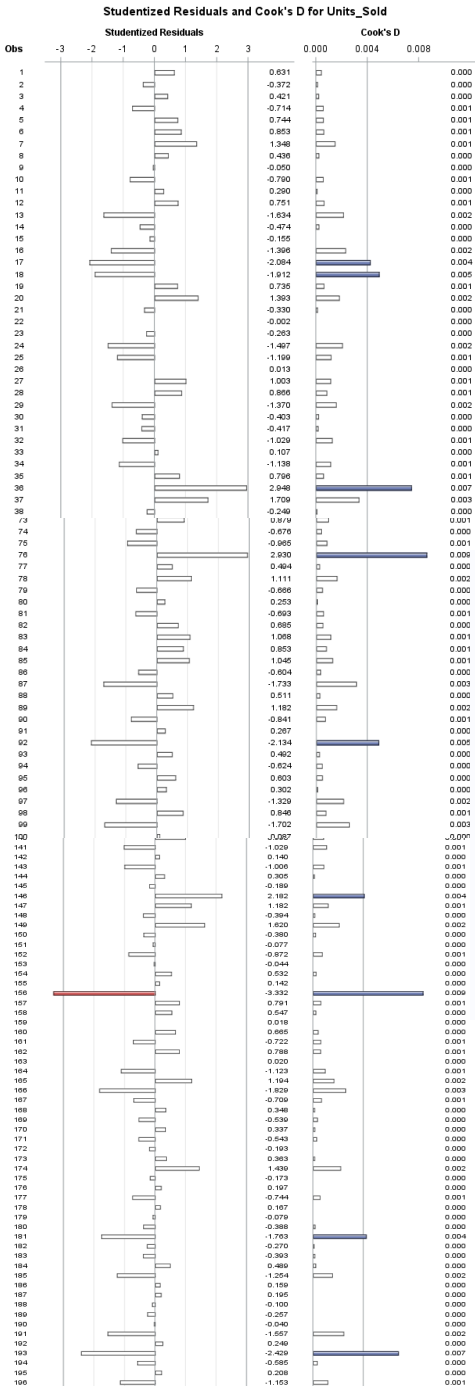
The REG Procedure  
Model: MODEL1  
Dependent Variable: Units\_Sold

|                             |      |
|-----------------------------|------|
| Number of Observations Read | 1000 |
| Number of Observations Used | 1000 |

| Analysis of Variance |     |                |             |         |        |
|----------------------|-----|----------------|-------------|---------|--------|
| Source               | DF  | Sum of Squares | Mean Square | F Value | Pr > F |
| Model                | 9   | 342.92880      | 38.10320    | 0.72    | 0.6914 |
| Error                | 990 | 52428          | 52.95749    |         |        |
| Corrected Total      | 999 | 52771          |             |         |        |

|                |          |          |         |
|----------------|----------|----------|---------|
| Root MSE       | 7.27719  | R-Square | 0.0065  |
| Dependent Mean | 29.63100 | Adj R-Sq | -0.0025 |
| Coeff Var      | 24.55938 |          |         |

| Parameter Estimates |    |                    |                |         |         |
|---------------------|----|--------------------|----------------|---------|---------|
| Variable            | DF | Parameter Estimate | Standard Error | t Value | Pr >  t |
| Intercept           | 1  | 29.00284           | 0.91763        | 31.61   | <.0001  |
| Price               | 1  | 0.00047564         | 0.00079713     | 0.60    | 0.5509  |
| Discount            | 1  | -0.01630           | 0.01607        | -1.01   | 0.3105  |
| Marketing_Spend     | 1  | 0.00002648         | 0.00008113     | 0.33    | 0.7442  |
| D_Premium           | 1  | 1.25551            | 0.57007        | 2.20    | 0.0279  |
| D_Regular           | 1  | 0.61756            | 0.55858        | 1.11    | 0.2692  |
| D_Sports            | 1  | 0.10379            | 0.71517        | 0.15    | 0.8846  |
| D_Toys              | 1  | -0.11071           | 0.71713        | -0.15   | 0.8773  |
| D_Fashion           | 1  | 0.19001            | 0.73029        | 0.26    | 0.7948  |
| D_HomeDecor         | 1  | 0.10174            | 0.73094        | 0.14    | 0.8893  |



This part checks for outliers and influential points in my model. Outliers are data points that are very different from the rest — they don't follow the general pattern. In my results, a few points like 156, 236, 310, 114, and 174 stand out because their actual sales were very different from what the model predicted. Influential points are data points that can strongly affect the regression line. Using Cook's Distance, I found points like 236, 310, and 274 that have slightly higher values, meaning they have some influence on the model. Overall, there are a few outliers and influential points, but none of them are serious or harmful to the

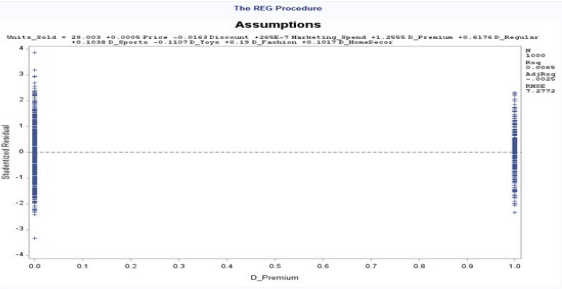
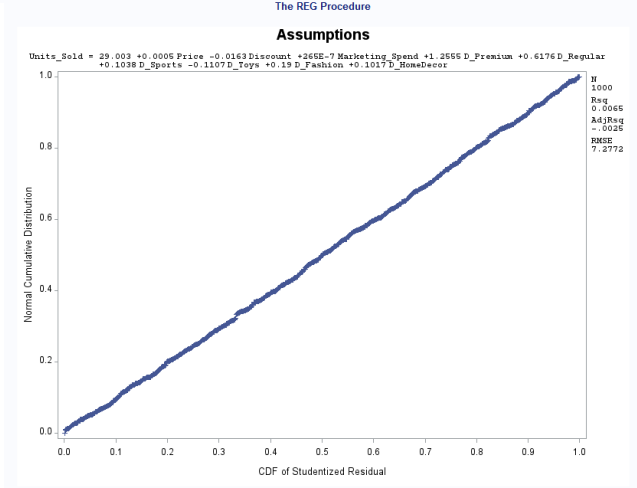
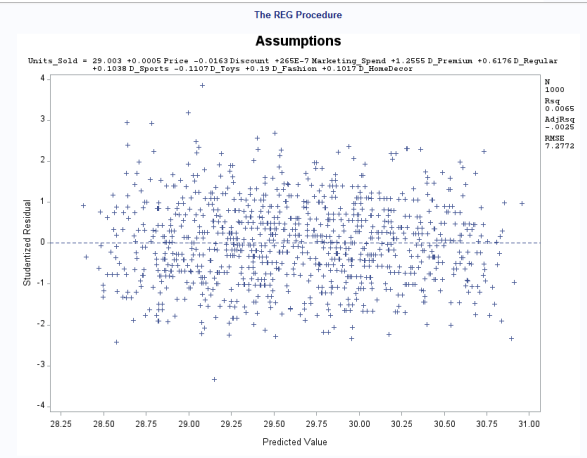
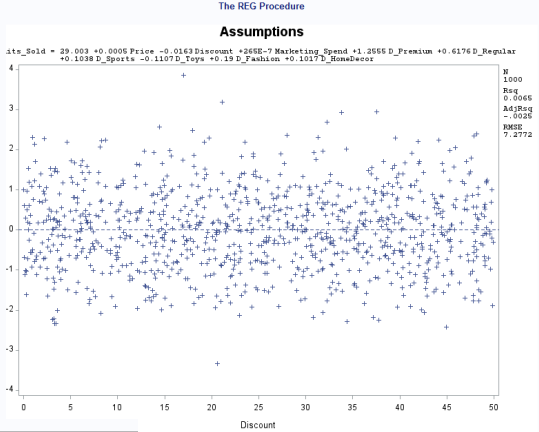
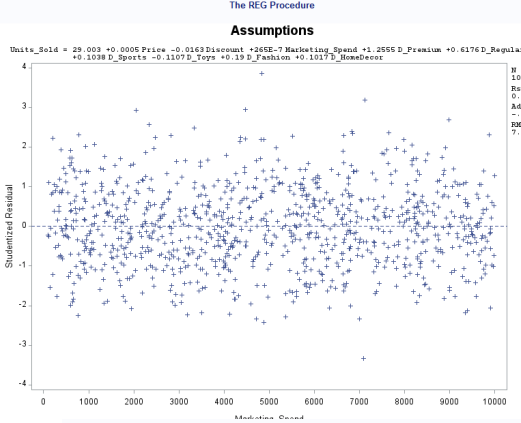
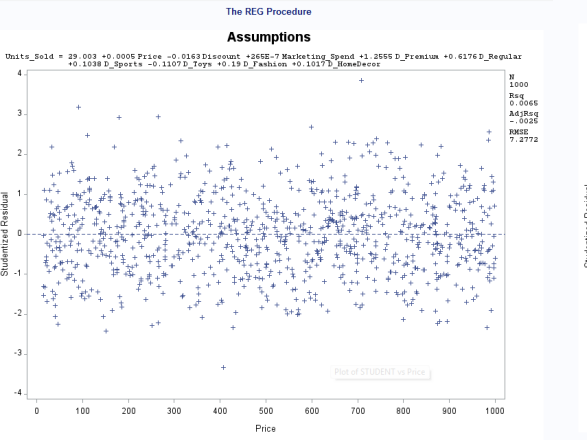
# ASSUMPTIONS

| The REG Procedure              |      |
|--------------------------------|------|
| Model: MODEL1                  |      |
| Dependent Variable: Units_Sold |      |
| Number of Observations Read    | 1000 |
| Number of Observations Used    | 1000 |

| Analysis of Variance |     |                |             |         |        |
|----------------------|-----|----------------|-------------|---------|--------|
| Source               | DF  | Sum of Squares | Mean Square | F Value | Pr > F |
| Model                | 9   | 342.92880      | 38.10320    | 0.72    | 0.6914 |
| Error                | 990 | 52428          | 52.95749    |         |        |
| Corrected Total      | 999 | 52771          |             |         |        |

|                |          |          |         |
|----------------|----------|----------|---------|
| Root MSE       | 7.27719  | R-Square | 0.0065  |
| Dependent Mean | 29.63100 | Adj R-Sq | -0.0025 |
| Coeff Var      | 24.55938 |          |         |

| Parameter Estimates |    |                    |                |         |         |
|---------------------|----|--------------------|----------------|---------|---------|
| Variable            | DF | Parameter Estimate | Standard Error | t Value | Pr >  t |
| Intercept           | 1  | 29.00284           | 0.91763        | 31.61   | <.0001  |
| Price               | 1  | 0.00047564         | 0.00079713     | 0.60    | 0.5509  |
| Discount            | 1  | -0.01630           | 0.01607        | -1.01   | 0.3105  |
| Marketing_Spend     | 1  | 0.00002648         | 0.00008113     | 0.33    | 0.7442  |
| D_Premium           | 1  | 1.25551            | 0.57007        | 2.20    | 0.0279  |
| D_Regular           | 1  | 0.61756            | 0.55858        | 1.11    | 0.2692  |
| D_Sports            | 1  | 0.10379            | 0.71517        | 0.15    | 0.8846  |
| D_Toys              | 1  | -0.11071           | 0.71713        | -0.15   | 0.8773  |
| D_Fashion           | 1  | 0.19001            | 0.73029        | 0.26    | 0.7948  |
| D_HomeDecor         | 1  | 0.10174            | 0.73094        | 0.14    | 0.8893  |



(ALL DUMMY  
VARIABLES ARE  
LIKE THIS)

This slide shows the regression assumptions for my model. First, I checked the residual plots for Price, Discount, and Marketing Spend. The dots are spread randomly above and below the zero line, this means the model meets the linearity and constant variance assumptions. There's no clear pattern or funnel shape, so that's a good sign. The next graph, the Normal Q-Q plot, shows that the points form a straight diagonal line. This means the residuals follow a normal distribution, which is another key assumption of regression. For all the dummy variables, the plots look the same, two straight vertical lines, because dummy variables only take values of 0 or 1, and that's completely normal. So overall, my model satisfies the main



After Removing Outliers and Influential Points

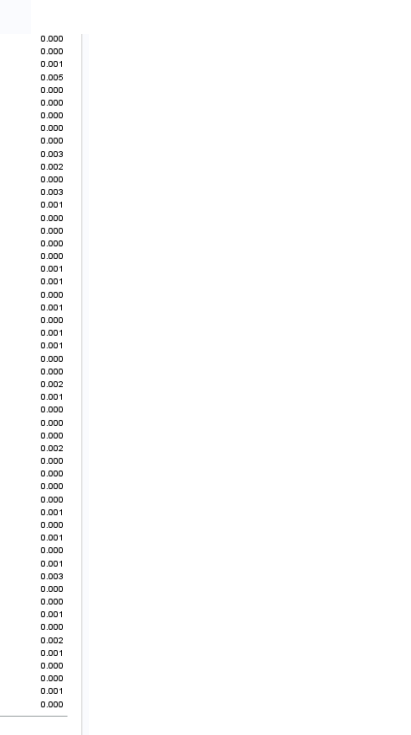
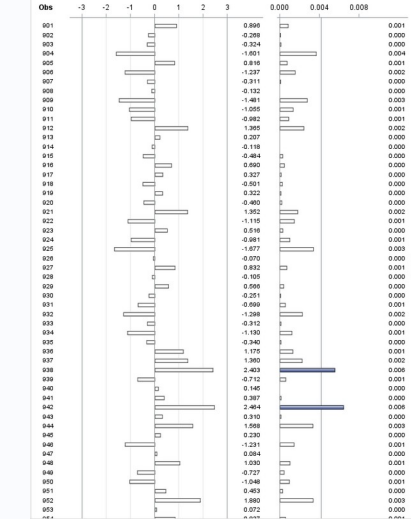
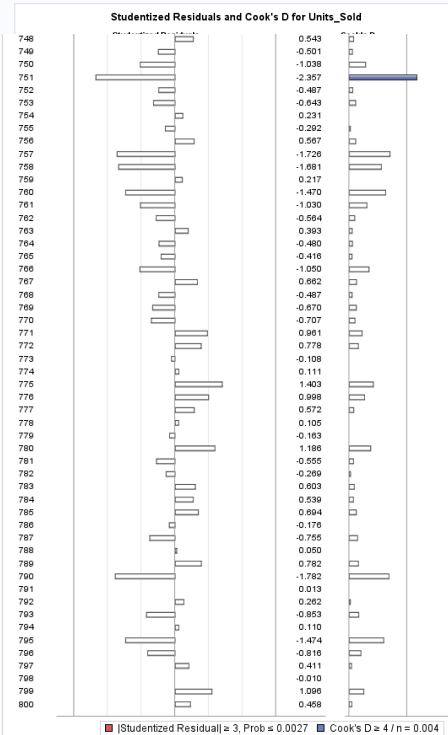
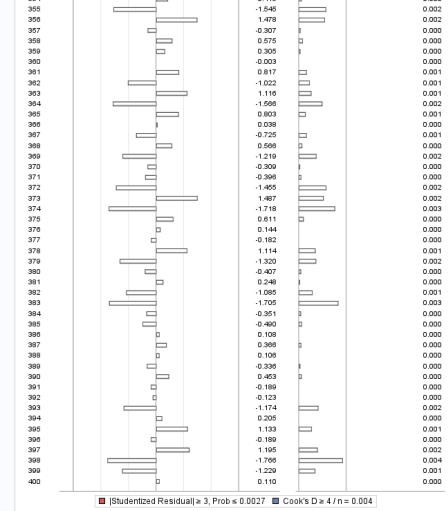
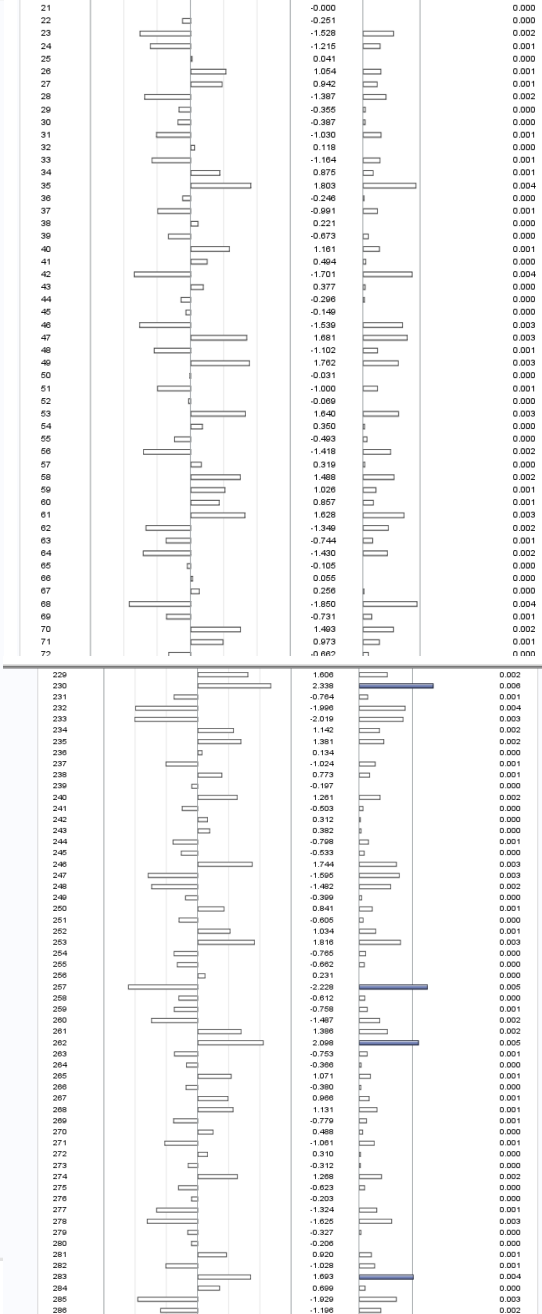
The REG Procedure  
Model: MODEL1  
Dependent Variable: Units\_Sold

|                             |     |
|-----------------------------|-----|
| Number of Observations Read | 955 |
| Number of Observations Used | 955 |

| Analysis of Variance |     |                |             |         |        |
|----------------------|-----|----------------|-------------|---------|--------|
| Source               | DF  | Sum of Squares | Mean Square | F Value | Pr > F |
| Model                | 9   | 450.43714      | 50.04857    | 1.02    | 0.4179 |
| Error                | 945 | 46152          | 48.83772    |         |        |
| Corrected Total      | 954 | 46602          |             |         |        |

|                |          |          |        |
|----------------|----------|----------|--------|
| Root MSE       | 6.98840  | R-Square | 0.0097 |
| Dependent Mean | 29.55288 | Adj R-Sq | 0.0002 |
| Coeff Var      | 23.64710 |          |        |

| Parameter Estimates |    |                    |                |         |         |
|---------------------|----|--------------------|----------------|---------|---------|
| Variable            | DF | Parameter Estimate | Standard Error | t Value | Pr >  t |
| Intercept           | 1  | 28.48419           | 0.90266        | 31.56   | <.0001  |
| Price               | 1  | 0.00070963         | 0.00078232     | 0.91    | 0.3646  |
| Discount            | 1  | -0.01303           | 0.01578        | -0.83   | 0.4091  |
| Marketing_Spend     | 1  | 0.00004387         | 0.00007987     | 0.55    | 0.5830  |
| D_Premium           | 1  | 1.35367            | 0.55949        | 2.42    | 0.0157  |
| D_Regular           | 1  | 0.80940            | 0.55175        | 1.47    | 0.1427  |
| D_Sports            | 1  | 0.21499            | 0.70001        | 0.31    | 0.7588  |
| D_Toys              | 1  | -0.32621           | 0.70178        | -0.46   | 0.6422  |
| D_Fashion           | 1  | 0.13724            | 0.71721        | 0.19    | 0.8483  |
| D_HomeDecor         | 1  | 0.54562            | 0.71761        | 0.76    | 0.4472  |



Final Model Assumption Checks – Cleaned Data

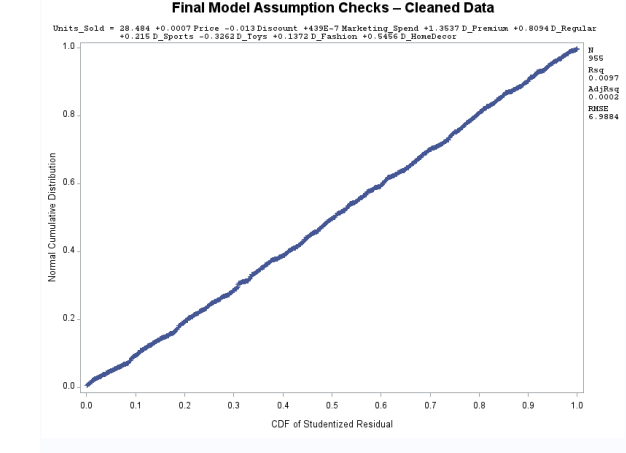
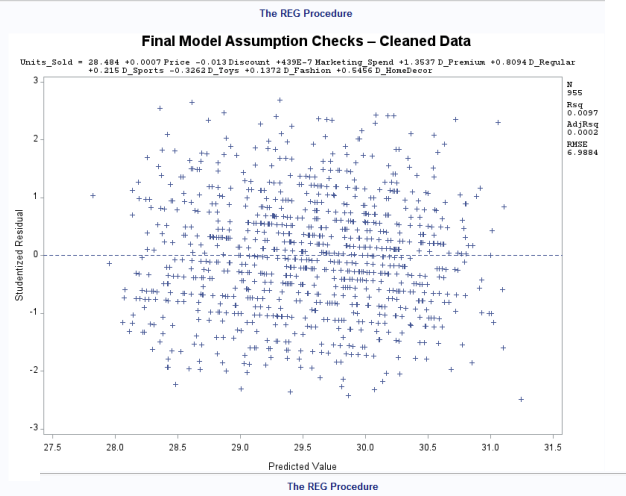
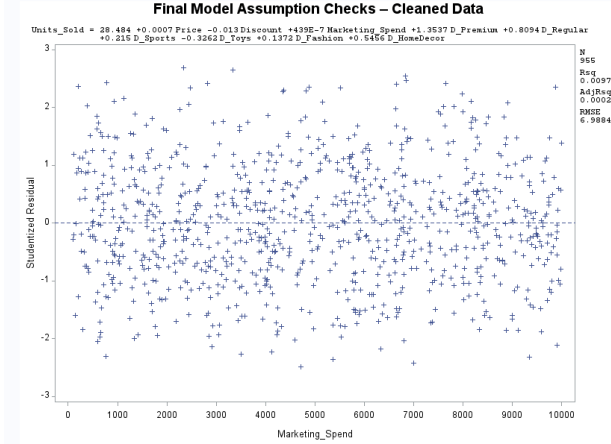
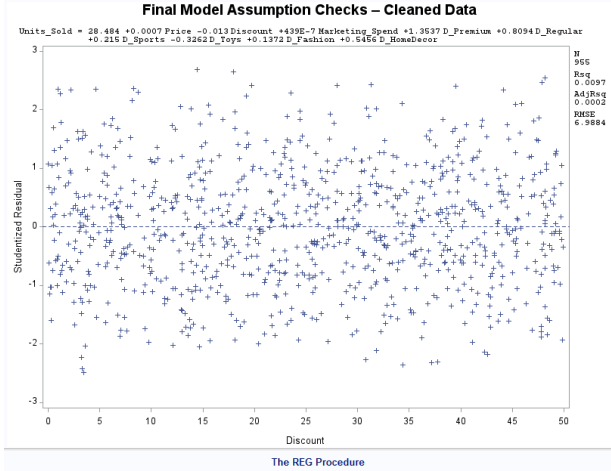
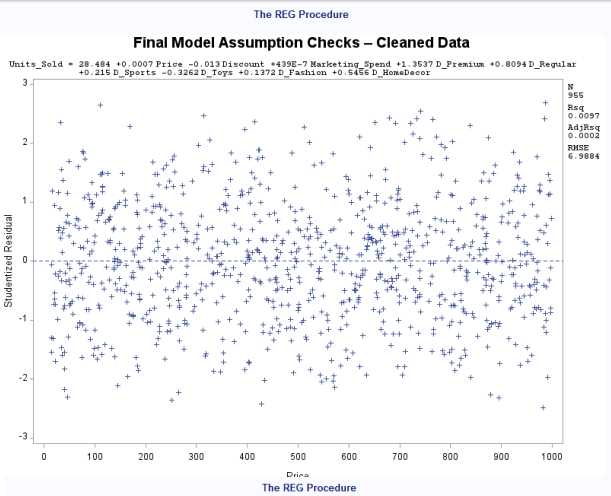
The REG Procedure  
Model: MODEL1  
Dependent Variable: Units\_Sold

|                             |     |
|-----------------------------|-----|
| Number of Observations Read | 955 |
| Number of Observations Used | 955 |

| Analysis of Variance |     |                |             |         |        |
|----------------------|-----|----------------|-------------|---------|--------|
| Source               | DF  | Sum of Squares | Mean Square | F Value | Pr > F |
| Model                | 9   | 450.43714      | 50.04857    | 1.02    | 0.4179 |
| Error                | 945 | 46152          | 48.83772    |         |        |
| Corrected Total      | 954 | 46602          |             |         |        |

|                |          |          |        |
|----------------|----------|----------|--------|
| Root MSE       | 6.98840  | R-Square | 0.0097 |
| Dependent Mean | 29.55288 | Adj R-Sq | 0.0002 |
| Coeff Var      | 23.64710 |          |        |

| Parameter Estimates |    |                    |                |         |         |
|---------------------|----|--------------------|----------------|---------|---------|
| Variable            | DF | Parameter Estimate | Standard Error | t Value | Pr >  t |
| Intercept           | 1  | 28.48419           | 0.90266        | 31.56   | <.0001  |
| Price               | 1  | 0.00070963         | 0.00078232     | 0.91    | 0.3646  |
| Discount            | 1  | -0.01303           | 0.01578        | -0.83   | 0.4091  |
| Marketing_Spend     | 1  | 0.00004387         | 0.00007987     | 0.55    | 0.5830  |
| D_Premium           | 1  | 1.35367            | 0.55949        | 2.42    | 0.0157  |
| D_Regular           | 1  | 0.80940            | 0.55175        | 1.47    | 0.1427  |
| D_Sports            | 1  | 0.21499            | 0.70001        | 0.31    | 0.7588  |
| D_Toy               | 1  | -0.32621           | 0.70178        | -0.46   | 0.6422  |
| D_Fashion           | 1  | 0.13724            | 0.71721        | 0.19    | 0.8483  |
| D_HomeDecor         | 1  | 0.54562            | 0.71761        | 0.76    | 0.4472  |



After I removed the outliers and influential points, I ran the regression again with 955 data points instead of 1000. The R-square went up a little from 0.0065 to 0.0097, which means the model explains a bit more variation, but still not much. The F-value increased from 0.72 to 1.02, and the p-value is still above 0.05, so the model is not statistically significant yet. However, removing the outliers made the model cleaner — the Root MSE dropped from 7.27 to 6.98, meaning the prediction errors are



# Test and Train Sets for Ecommerce Data

## The SURVEYSELECT Procedure

|                  |                        |
|------------------|------------------------|
| Selection Method | Simple Random Sampling |
|------------------|------------------------|

|                       |                 |
|-----------------------|-----------------|
| Input Data Set        | ECommerce_CLEAN |
| Random Number Seed    | 495857          |
| Sampling Rate         | 0.75            |
| Sample Size           | 717             |
| Selection Probability | 0.750785        |
| Sampling Weight       | 0               |
| Output Data Set       | XV_ALL          |

In this step, I split my cleaned dataset, called `ecommerce_clean`, into two parts — training and testing sets — using random sampling. I used a sampling rate of 0.75, which means 75% of the data, or 717 observations, were used for training, and the remaining 25% for testing. I also used a random seed of 495857 so that if I run the code again, I get the same split. The output dataset, called `XV_ALL`, includes both parts and shows

| Creating new Y |          |            |                  |        |          |                  |                 |            |           |           |          |        |           |             |       |
|----------------|----------|------------|------------------|--------|----------|------------------|-----------------|------------|-----------|-----------|----------|--------|-----------|-------------|-------|
| Obs            | Selected | Date       | Product_Category | Price  | Discount | Customer_Segment | Marketing_Spend | Units_Sold | D_Premium | D_Regular | D_Sports | D_Toys | D_Fashion | D_HomeDecor | new_Y |
| 1              | 1        | 25/01/2023 | Electronics      | 804.82 | 10.28    | Occasional       | 8047.83         | 34         | 0         | 0         | 0        | 0      | 0         | 0           | 34    |
| 2              | 0        | 08/02/2023 | Electronics      | 503.38 | 40.59    | Occasional       | 4291.02         | 26         | 0         | 0         | 0        | 0      | 0         | 0           | .     |
| 3              | 1        | 25/02/2023 | Electronics      | 752.46 | 40.8     | Occasional       | 9736.49         | 32         | 0         | 0         | 0        | 0      | 0         | 0           | 32    |
| 4              | 1        | 12/03/2023 | Electronics      | 163.69 | 2.16     | Occasional       | 4744.16         | 24         | 0         | 0         | 0        | 0      | 0         | 0           | 24    |
| 5              | 0        | 29/03/2023 | Electronics      | 113.07 | 38.06    | Occasional       | 6772.54         | 34         | 0         | 0         | 0        | 0      | 0         | 0           | .     |
| 6              | 1        | 16/04/2023 | Electronics      | 216.81 | 27.71    | Occasional       | 6277.59         | 36         | 0         | 0         | 0        | 0      | 0         | 0           | 36    |
| 7              | 1        | 25/04/2023 | Electronics      | 594.39 | 16.28    | Occasional       | 7277.56         | 39         | 0         | 0         | 0        | 0      | 0         | 0           | 39    |
| 8              | 1        | 21/05/2023 | Electronics      | 28.21  | 13.49    | Occasional       | 1837.37         | 32         | 0         | 0         | 0        | 0      | 0         | 0           | 32    |
| 9              | 0        | 01/06/2023 | Electronics      | 901.02 | 7.37     | Occasional       | 1934.18         | 29         | 0         | 0         | 0        | 0      | 0         | 0           | .     |
| 10             | 1        | 18/06/2023 | Electronics      | 296.81 | 28.59    | Occasional       | 1833.72         | 23         | 0         | 0         | 0        | 0      | 0         | 0           | 23    |
| 11             | 1        | 27/06/2023 | Electronics      | 222.79 | 25.9     | Occasional       | 7946.69         | 31         | 0         | 0         | 0        | 0      | 0         | 0           | 31    |
| 12             | 1        | 29/06/2023 | Electronics      | 269.64 | 47.03    | Occasional       | 7527.63         | 34         | 0         | 0         | 0        | 0      | 0         | 0           | 34    |
| 13             | 0        | 26/08/2023 | Electronics      | 237.71 | 23.2     | Occasional       | 3934.49         | 17         | 0         | 0         | 0        | 0      | 0         | 0           | .     |
| 14             | 1        | 30/08/2023 | Electronics      | 727.52 | 7.56     | Occasional       | 7774.92         | 26         | 0         | 0         | 0        | 0      | 0         | 0           | 26    |
| 15             | 1        | 21/09/2023 | Electronics      | 631.23 | 27.27    | Occasional       | 9952.08         | 28         | 0         | 0         | 0        | 0      | 0         | 0           | 28    |
| 16             | 1        | 29/09/2023 | Electronics      | 119.37 | 12.93    | Occasional       | 9361.3          | 19         | 0         | 0         | 0        | 0      | 0         | 0           | 19    |
| 17             | 1        | 20/10/2023 | Electronics      | 991.26 | 41.11    | Occasional       | 650.49          | 16         | 0         | 0         | 0        | 0      | 0         | 0           | 16    |
| 18             | 1        | 27/10/2023 | Electronics      | 184.76 | 39.64    | Occasional       | 8966.07         | 34         | 0         | 0         | 0        | 0      | 0         | 0           | 34    |
| 19             | 1        | 30/10/2023 | Electronics      | 781.79 | 39.9     | Occasional       | 6941.46         | 39         | 0         | 0         | 0        | 0      | 0         | 0           | 39    |
| 20             | 1        | 03/11/2023 | Electronics      | 884.96 | 7.9      | Occasional       | 3486.87         | 27         | 0         | 0         | 0        | 0      | 0         | 0           | 27    |
| 21             | 0        | 21/11/2023 | Electronics      | 889.86 | 40.18    | Occasional       | 9322.39         | 29         | 0         | 0         | 0        | 0      | 0         | 0           | .     |
| 22             | 0        | 18/01/2024 | Electronics      | 763.42 | 34.71    | Occasional       | 3907.66         | 27         | 0         | 0         | 0        | 0      | 0         | 0           | .     |
| 23             | 1        | 29/01/2024 | Electronics      | 711.09 | 33.26    | Occasional       | 1696.07         | 18         | 0         | 0         | 0        | 0      | 0         | 0           | 18    |
| 24             | 1        | 19/02/2024 | Electronics      | 431.92 | 37.7     | Occasional       | 3613.49         | 20         | 0         | 0         | 0        | 0      | 0         | 0           | 20    |
| 25             | 1        | 07/03/2024 | Electronics      | 281.47 | 29.92    | Occasional       | 9686.37         | 29         | 0         | 0         | 0        | 0      | 0         | 0           | 29    |
| 26             | 1        | 23/03/2024 | Electronics      | 846.34 | 49.77    | Occasional       | 5448.94         | 36         | 0         | 0         | 0        | 0      | 0         | 0           | 36    |
| 27             | 1        | 13/04/2024 | Electronics      | 21.5   | 30.36    | Occasional       | 8073.76         | 36         | 0         | 0         | 0        | 0      | 0         | 0           | 36    |

In this step, I created a new variable called `new_Y` to store the dependent variable (`Units_Sold`) only for the training data.

Backward selection is a method used in regression to find which variables really matter. It starts with all the predictors in the model and then removes the least useful ones step by step

#### Backward Elimination: Step 0

All Variables Entered: R-Square = 0.0140 and C(p) = 10.0000

| Analysis of Variance |     |                |             |         |        |
|----------------------|-----|----------------|-------------|---------|--------|
| Source               | DF  | Sum of Squares | Mean Square | F Value | Pr > F |
| Model                | 9   | 480.22436      | 53.35826    | 1.11    | 0.3509 |
| Error                | 707 | 33898          | 47.94559    |         |        |
| Corrected Total      | 716 | 34378          |             |         |        |

| Variable        | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|-----------------|--------------------|----------------|------------|---------|--------|
| Intercept       | 28.33586           | 1.04562        | 35211      | 734.38  | <.0001 |
| Price           | 0.00097629         | 0.00091079     | 55.08996   | 1.15    | 0.2841 |
| Discount        | -0.00258           | 0.01816        | 0.97164    | 0.02    | 0.8868 |
| Marketing_Spend | 0.00007054         | 0.00009060     | 29.05967   | 0.61    | 0.4365 |
| D_Premium       | 1.73219            | 0.64549        | 345.27667  | 7.20    | 0.0075 |
| D_Regular       | 0.90857            | 0.62775        | 100.43614  | 2.09    | 0.1482 |
| D_Sports        | -0.28509           | 0.80271        | 6.04760    | 0.13    | 0.7226 |
| D_Toys          | -0.71783           | 0.81324        | 37.35535   | 0.78    | 0.3777 |
| D_Fashion       | -0.68822           | 0.82344        | 33.49227   | 0.70    | 0.4036 |
| D_HomeDecor     | -0.14333           | 0.82272        | 1.45513    | 0.03    | 0.8617 |

| Analysis of Variance |     |                |             |         |        |
|----------------------|-----|----------------|-------------|---------|--------|
| Source               | DF  | Sum of Squares | Mean Square | F Value | Pr > F |
| Model                | 4   | 413.21318      | 103.30329   | 2.17    | 0.0713 |
| Error                | 712 | 33965          | 47.70301    |         |        |
| Corrected Total      | 716 | 34378          |             |         |        |

| Variable  | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|-----------|--------------------|----------------|------------|---------|--------|
| Intercept | 28.34505           | 0.66750        | 86019      | 1803.23 | <.0001 |
| Price     | 0.00093805         | 0.00090509     | 51.24048   | 1.07    | 0.3004 |
| D_Premium | 1.75131            | 0.64291        | 353.97638  | 7.42    | 0.0066 |
| D_Regular | 0.93502            | 0.62554        | 106.58117  | 2.23    | 0.1354 |
| D_Toys    | -0.43223           | 0.64620        | 21.34227   | 0.45    | 0.5038 |

Bounds on condition number: 1.3473, 18.793

#### Backward Elimination: Step 6

Variable D\_Toys Removed: R-Square = 0.0114 and C(p) = -0.1572

| Analysis of Variance |     |                |             |         |        |
|----------------------|-----|----------------|-------------|---------|--------|
| Source               | DF  | Sum of Squares | Mean Square | F Value | Pr > F |
| Model                | 3   | 391.87090      | 130.62363   | 2.74    | 0.0424 |
| Error                | 713 | 33986          | 47.66604    |         |        |
| Corrected Total      | 716 | 34378          |             |         |        |

| Variable  | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|-----------|--------------------|----------------|------------|---------|--------|
| Intercept | 28.24631           | 0.65072        | 89813      | 1884.22 | <.0001 |
| Price     | 0.00095181         | 0.00090451     | 52.78224   | 1.11    | 0.2930 |
| D_Premium | 1.75257            | 0.64266        | 354.48646  | 7.44    | 0.0065 |
| D_Regular | 0.94988            | 0.62490        | 110.13537  | 2.31    | 0.1289 |

#### Backward Elimination: Step 8

Variable D\_Regular Removed: R-Square = 0.0067 and C(p) = -0.8141

| Analysis of Variance |     |                |             |         |        |
|----------------------|-----|----------------|-------------|---------|--------|
| Source               | DF  | Sum of Squares | Mean Square | F Value | Pr > F |
| Model                | 1   | 231.58522      | 231.58522   | 4.85    | 0.0280 |
| Error                | 715 | 34146          | 47.75688    |         |        |
| Corrected Total      | 716 | 34378          |             |         |        |

| Variable  | Parameter Estimate | Standard Error | Type II SS | F Value | Pr > F |
|-----------|--------------------|----------------|------------|---------|--------|
| Intercept | 29.22699           | 0.31251        | 417712     | 8746.64 | <.0001 |
| D_Premium | 1.22037            | 0.55419        | 231.58522  | 4.85    | 0.0280 |

This slide shows the backward selection process. Backward selection is a method that helps find which variables are really important in explaining sales. It starts with all predictors in the model and then removes the least useful ones step by step, based on their p-values. At first, the R-square was 0.0140, meaning the model explained only about 1.4% of the variation. As we removed the variables with high p-values, the model kept getting simpler. In the final step, only the variable D\_Premium remained, which means Premium customers have a significant positive impact on Units Sold. The final R-square is 0.0067 with a p-value of 0.028, showing that only this variable is statistically meaningful. So, backward selection

Validation (train set)

The REG Procedure  
Model: MODEL1  
Dependent Variable: new\_Y

|                                            |     |
|--------------------------------------------|-----|
| Number of Observations Read                | 955 |
| Number of Observations Used                | 717 |
| Number of Observations with Missing Values | 238 |

Analysis of Variance

| Source          | DF  | Sum of Squares | Mean Square | F Value | Pr > F |
|-----------------|-----|----------------|-------------|---------|--------|
| Model           | 1   | 231.58522      | 231.58522   | 4.85    | 0.0280 |
| Error           | 715 | 34146          | 47.75688    |         |        |
| Corrected Total | 716 | 34378          |             |         |        |

|                |          |          |        |
|----------------|----------|----------|--------|
| Root MSE       | 6.91064  | R-Square | 0.0067 |
| Dependent Mean | 29.61506 | Adj R-Sq | 0.0053 |
| Coeff Var      | 23.33487 |          |        |

Parameter Estimates

| Variable  | DF | Parameter Estimate | Standard Error | t Value | Pr >  t |
|-----------|----|--------------------|----------------|---------|---------|
| Intercept | 1  | 29.22699           | 0.31251        | 93.52   | <.0001  |
| D_Premium | 1  | 1.22037            | 0.55419        | 2.20    | 0.0280  |

The Root MSE is about 6.9, meaning that on average, the model’s predictions are off by around 7 units. The only significant variable is D\_Premium, showing that Premium customers buy a bit more than others.

On the left side, we have the regression output from the training set. The R-square value is 0.0067, which means the model explains about 0.67% of the variation in sales — so it’s quite low, but the model is still statistically significant.

| Difference Between Observed and Predicted in Test Set |            |                      |          |         |
|-------------------------------------------------------|------------|----------------------|----------|---------|
| Obs                                                   | Units_Sold | Predicted_Units_Sold | d        | absd    |
| 1                                                     | 26         | 29.2270              | -3.2270  | 3.2270  |
| 2                                                     | 34         | 29.2270              | 4.7730   | 4.7730  |
| 3                                                     | 29         | 29.2270              | -0.2270  | 0.2270  |
| 4                                                     | 17         | 29.2270              | -12.2270 | 12.2270 |
| 5                                                     | 29         | 29.2270              | -0.2270  | 0.2270  |
| 6                                                     | 27         | 29.2270              | -2.2270  | 2.2270  |
| 7                                                     | 19         | 29.2270              | -10.2270 | 10.2270 |
| 8                                                     | 30         | 29.2270              | 0.7730   | 0.7730  |
| 9                                                     | 22         | 29.2270              | -7.2270  | 7.2270  |
| 10                                                    | 37         | 29.2270              | 7.7730   | 7.7730  |
| 11                                                    | 17         | 29.2270              | -12.2270 | 12.2270 |
| 12                                                    | 18         | 29.2270              | -11.2270 | 11.2270 |
| 13                                                    | 22         | 29.2270              | -7.2270  | 7.2270  |
| 14                                                    | 40         | 29.2270              | 10.7730  | 10.7730 |
| 15                                                    | 19         | 29.2270              | -10.2270 | 10.2270 |
| 16                                                    | 31         | 29.2270              | 1.7730   | 1.7730  |
| 17                                                    | 39         | 29.2270              | 9.7730   | 9.7730  |
| 18                                                    | 23         | 29.2270              | -6.2270  | 6.2270  |
| 19                                                    | 29         | 29.2270              | -0.2270  | 0.2270  |

On the right side, we can see the comparison between the actual Units Sold and the predicted Units Sold in the test set. The differences (or errors) are small, which means the model is predicting consistently, even though the overall variation explained is low. In short, the model performs okay, it’s clean, stable, and confirms that Premium customers tend to buy slightly more units.